Status	Finished
Started	Monday, 5 October 2026, 7:14 PM
Completed	Saturday, 17 October 2026, 5:23 PM
Duration	11 days 22 hours
Marks	3.00/3.00
Grade	**3.75** out of 3.75 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Discuss the following system of linear equations:

$$\begin{pmatrix} 1 & 0 & 6 \\ 5 & 8 & -9 \\ -3 & 5 & 6 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ -6 \\ -3 \end{pmatrix}.$$

Which of the following statements is true?

a. Consistent and independent with trivial solution
b. Inconsistent
c. Consistent and dependent with 1 degree of freedom
d. Consistent and dependent with 2 degrees of freedom
e. Consistent and independent ✓

To find the number of solutions of the system of equations, we need to know the rank of the coefficient matrix and of its augmented matrix.

In this case, the augmented matrix of the system is:

$$\begin{pmatrix} 1 & 0 & 6 & 1 \\ 5 & 8 & -9 & -6 \\ -3 & 5 & 6 & -3 \end{pmatrix}.$$

To find the rank of both matrices, we can either use Gaussian elimination or compute the rank using minors.

If we do Gaussian elimination, we get

$$\begin{pmatrix} 1 & 0 & 6 & 1 \\ 0 & 8 & -39 & -11 \\ 0 & 0 & \frac{387}{8} & \frac{55}{8} \end{pmatrix}.$$

In this case, since the rank of the coefficient matrix is 3 and the rank of the augmented matrix is 3 the correct answer is: Consistent and independent.

Remember that:

- a system of equations is inconsistent if the rank of the coefficient matrix and of the augmented matrix are different

- a system of equations is consistent and independent (admits a unique solution) if the rank of the coefficient matrix and of the augmented matrix are equal, and they coincide with the number of variables (three in this case)

- a system of equations is consistent and dependent (the system has infinitely many solutions) if the rank of the coefficient matrix and of the augmented matrix are equal, but smaller than the number of variables (the difference between the number of variables and the rank is called the number of degrees of freedom)

- a system of equations is consistent and independent with trivial solution if the unique solution of the system is the zero solution (in this case, if the coefficient matrix has rank 3 and the system is homogeneous - zero constant vector).

The correct answer is: Consistent and independent

Question 2

Correct

Mark 1.00 out of 1.00

Express the following system of equations in matrix notation:

$$21\cdot x+4\cdot y-8\cdot z+10=0,$$
$$-9\cdot x+27\cdot y-10\cdot z-9=0,$$
$$-7\cdot x+4\cdot y+16\cdot z-3=0,$$

taking into account that the vector of variables is (x,y,z).

coefficient matrix =

21	4	-8
-9	27	-10
-7	4	16

constant vector =

-10
9
3

augmented matrix =

21	4	-8	-10
-9	27	-10	9
-7	4	16	3

The first thing we need to do is to move all the terms with variables to the left and all constant to the right, yielding the system:

$$21\cdot x+4\cdot y-8\cdot z=-10,$$
$$-9\cdot x+27\cdot y-10\cdot z=9,$$
$$-7\cdot x+4\cdot y+16\cdot z=3,$$

Taking into account that the vector containing the variables is (x,y,z), we obtain the coefficient matrix by stacking the x's $(21,-9,-7)$, y's $(4,27,4)$ and z's $(-8,-10,16)$ coefficients columnwise.

Therefore, the coefficient matrix is $A=\begin{pmatrix}21&4&-8\\-9&27&-10\\-7&4&16\end{pmatrix}$.

The constant vector is built using the constants that appear on the right-hand side of the equal sign in each equation $(-10,9,3)$.

That is, $b=\begin{pmatrix}-10\\9\\3\end{pmatrix}$.

Finally, the augmented matrix of the system is the matrix obtained concatenating the matrix (A) and the constant vector (b). That is,

the augmented matrix is $(A|b)=\left(\begin{array}{cccc}21&4&-8&-10\\-9&27&-10&9\\-7&4&16&3\end{array}\right)$.

Question 3

Correct

Mark 1.00 out of 1.00

Consider the following system of equations given in matrix form

$$\begin{pmatrix} 1 & 5\cdot a-30 & 3\cdot a \\ 0 & a-6 & a \\ 0 & 9\cdot a-54 & 10\cdot a \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix}=\begin{pmatrix} 8\cdot a-53 \\ 2\cdot a-12 \\ 19\cdot a-114 \end{pmatrix},$$

where $a \in \mathbb{R}$ is a parameter.

Which values of a yield to a consistent and dependent system of equations? How many degrees of freedom does the system have in this case?

a = 6 ✓

degrees of freedom = 1 ✓

What value of a yields to an inconsistent system of equations?

a = 0 ✓

FORMAT OF THE ANSWER: If any of the results is a fraction, we have to write it down as a fraction and not as a decimal. For example, 15/7, -1/21.

To determine the number of solutions of the system, we will use the Rouché-Capelli theorem. Therefore, we need to know the rank of the coefficient and augmented matrix.

These ranks can be obtained by:

(1) using Gaussian elimination to the augmented matrix

(2) computing determinants

Since in this case we don't have to find the solutions of the system of equations, it can be faster to use determinants. In this case, we have that

$\det(A)=a^2-6\cdot a = 0 \iff a \in \{0, 6\}$

Therefore,

- if $a \notin \{0, 6\}$, rank(A)=3=number of variables and the system is consistent and independent
- $a = 6$

  In this case, the augmented matrix of the system is $(A|b)=\begin{pmatrix} 1 & 0 & 18 & -5 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 60 & 0 \end{pmatrix}$.

  Therefore, we have a consistent and dependent system with one degree of freedom, since rank(A)=rank(A|b)=2<3.
- $a = 0$

  In this case, the augmented matrix of the system is $(A|b)=\begin{pmatrix} 1 & -30 & 0 & -53 \\ 0 & -6 & 0 & -12 \\ 0 & -54 & 0 & -114 \end{pmatrix}$.

  Therefore, we have an inconsistent system since the second and third equations are incompatible: rank(A)=2 and rank(A|b)=3